Acta Materialia 153 (2018) 327–335

Contents lists available at ScienceDirect

Acta Materialia

journal homepage: www.elsevier.com/locate/actamat

Full length article

Effect of solutes on ideal shear resistance and electronic properties of magnesium: A first-principles study

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ARTICLE INFO

Article history:
Received 7 February 2018
Received in revised form
18 April 2018
Accepted 7 May 2018
Available online 9 May 2018

Keywords:
Magnesium
Ideal shear resistance
Electronic properties
First principles

ABSTRACT

Solution strengthening or softening is an effective way to enhance mechanical properties, especially in magnesium based alloys due to their inability to activate adequate non-basal deformation mechanisms at the room temperature. Hence, using first-principles calculations, the effects of several different alloying elements on the ideal shear resistance across various slip systems of Mg were investigated. The results reveal that the addition of a Ce or Zr solute atom decreases the ideal shear resistance (softening); whereas, the substitution of a Sn, Li or Zn atom increases the ideal shear resistance of Mg (strengthening). The dominant slip system in Mg was found to change from the basal partial $(0001)[10\bar{1}0]$ to prismatic $(10\bar{1}0)[11\bar{2}0]$ with the addition of a Ce or Zr solute atom; whereas, the addition of a Sn, Li or Zn solute atom had negligible effect on the plastic anisotropy. Furthermore, the electronic density of states and valence charge transfer, which provides a quantum mechanical insight into the underlying factors influencing the observed softening/strengthening behavior, was probed. For instance, the electronic density of states calculations show that the contribution from *d* states of Ce and Zr solute atoms decreases the electronic structure stability of their respective solid solution, thereby enhancing slip activities. In the end, theoretical analyses were performed and a shearability parameter was introduced to understand the implications of the observed variation in ideal shear resistance on the macroscopic behavior of Mg alloys.

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1. Introduction

The properties of metals change with addition of alloying elements and can be used to tailor or enhance their mechanical properties. This is specifically true for magnesium (Mg) alloys, which in today's modern industrial society, have been identified as a viable light-weight alternative to iron- and aluminum-based alloys for structural applications [1–4]. The adaptation of Mg and its alloys to large scale applications has been restricted due to their limited room temperature plasticity, that stems from the inability of Mg alloys to activate sufficient non-basal slip systems [5,6]. Thus, to improve the strength and formability of Mg at room temperature, the effects of various solute additions [7–12], especially rare earth (RE) elements like yttrium (Y), lanthanum (La) [13–17] etc. have been extensively studied. For instance, experimentally, enhanced room temperature ductility in a Mg-3 wt% Y alloy was reported due to the activation of the pyramidal <c+a> slip system [18]. Based on atomistic simulations, this behavior was attributed to an increase in the critical resolved shear stress (CRSS) of the basal slip system due to the addition of Y, thus the activation probabilities of basal and pyramidal slip systems become equal [19]. On the contrary, experimentally, strengthening of various slip systems and extension twinning was also observed upon Y addition in Mg [7]. In Mg-Al alloys, enhanced ductility was reported due to grain boundary sliding [20] as well as a decrease in the basal plane stacking fault energy [21,22]. However, strengthening upon addition of Al to pure Mg has been reported due to the formation of strong bonds between Al and Mg atoms [23,24]. The addition of solutes, such as Ce, Zn and Li, respectively, to Mg alloys was found to improve the room temperature ductility of Mg [14,25,26]. Further, in the case of Li and Zn solutes, the increase in ductility of Mg alloys was due to the activation of non-basal <c+a> and prismatic slip systems [14,25,26]. However, tensile tests on extruded Mg-Zn alloys show that the yield strength increases with Zn addition compared to pure Mg due to solid-solution strengthening in Mg-Zn alloys [27]. Such divergent experimental and theoretical

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https://doi.org/10.1016/j.actamat.2018.05.014
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results reflect the ambiguous understanding of the effects of different solute atoms on the plastic deformation in Mg.

Theoretical studies have also been utilized to gain valuable insights into the effect of solute addition or various defects on the plastic deformation of a material [28–31]. Computational studies have generally focused on calculating ideal strength (maximum attainable stress) and generalized stacking fault energy (GSFE) which closely relates to the dislocation properties and plasticity of the material [29,32,33]. The strength of a real crystal is affected by various defects like dislocations, grain boundaries etc. but the ideal shear strength/resistance for a defect free crystal offers fundamental suggestions, especially for the dislocation nucleation or shearability of the crystal, i.e., minimum shear stress required to instigate plastic deformation in a pure crystal with no defects [33]. Therefore, the study of the ideal strength of a defect-free crystal can provide vital information to understand the mechanical properties as it dictates the stress required for dislocation nucleation in a pristine crystal [34]. For instance, using first principles calculations, the ideal shear strength (resistance) of many metals and ceramics was quantified by Ogata et al. [30] and was found to be related to the valance charge distribution and directional bonding between the atoms. Furthermore, quantifying directly the effect of various solutes on the ideal shear resistance in Mg solid solutions will provide novel insights into the bonding behavior of solute and host metal atoms, and subsequent softening or hardening behavior.

In this work, first-principles calculations were performed to quantify the ideal shear resistance (τ) for pure Mg and Mg-X solid solutions (X = Li, Zn, Zr, Sn, Ce) along different slip systems to study the effect of solute addition on shearability. The effect of different solutes on the ideal shear resistance of Mg can be correlated to bonding between the solute and Mg atoms, which can be inferred from the electronic structure calculations for different Mg solid solutions; thus providing critical information on alloy design or screening. The changes in electronic structure of Mg due to the presence of a solute atom were revealed with the help of electronic density of states (DOS) and valence charge density distribution analysis. Furthermore, the energetic effect of solute addition was separated into mechanical (elastic) and chemical (electronic) contributions to comprehend the changes associated at the atomic length scale. Finally, to understand the macroscopic implications of these findings, the orientation dependence of active slip systems based on Schmid's Law and ideal shear resistance were computed. A shearability parameter was also introduced to estimate the strengthening or softening effect of solute atoms on different slip systems of Mg.

2. Computational technique

In the present study, the density functional theory (DFT) based Vienna ab initio simulation package (VASP) was used to examine the ideal shear resistance of Mg alloys [35,36]. The projector augmented wave (PAW) pseudopotentials [37,38] were used to represent Mg and various solute elements. The generalized gradient approximation (GGA) using the Perdew-Burke-Ernzerhof (PBE) [39] formulation were used to represent the exchange and correlation functions with an energy cutoff of 275 eV for Mg. Gamma pack k-point mesh of 16×9 x 10 was selected after extensive k-point sampling convergence studies to perform Brillouin-zone integrations. The ions were relaxed using the quasi-Newtonian method [40] with a force and energy convergence criteria of 1 meVÅ^{-1} and 10^{-6} eV, respectively. The computed equilibrium structure parameters like lattice constants, cohesive energy and elastic constants were found to match very well with the previously reported experimental and theoretical values [16,41,42]. For instance, the calculated equilibrium lattice parameters a = 3.192 Å and c/a = 1.623 agrees with the previous experimental and theoretical findings [43,44]. In the case of an Mg-X solid solution, the supercell was obtained from 3 periodic lengths of a pure Mg unit cell along the different orientations given in Table S1 (refer supplementary document). Then, one desired solute atom was substituted in the supercell to maintain a solute concentration of 0.93 at.%. The substituted solute atom was allowed to relax in all directions. Since, the simple shear deformation of the simulation cell was carried out with periodic boundary conditions; the ideal shear resistance calculations are independent of the position of the solute atom in the simulation cell. The self-interactions between the substitutional atoms for the chosen supercell were negligible. For instance, the binding energy of Zr in Mg with 108 atoms (i.e. a $3 \times 3 \times 3$ supercell) and 256 atoms (i.e., a $4 \times 4 \times 4$ supercell) was found to be 0.01 eV/atom. Therefore, to minimize the computational requirements, we chose the $3 \times 3 \times 3$ supercell of Mg for this work. Furthermore, the chosen supercell is in agreement with previous DFT studies to calculate generalized stacking fault energies [22,45–47] and ideal shear strength [30]. The cut-off energy and the k-point mesh density for the various solutes were selected based on convergence studies and the details can be found in Table S2 (refer supplementary document). The ideal shear resistance calculations were performed by applying simple shear strain along the slip direction in increments of 1% until yielding occurred. The atomic positions and basis vectors were fully relaxed at each step while keeping the applied strain fixed. Next, the electronic density of states for the unstrained configurations and valence charge density distribution were calculated to understand the variations in electronic structure of the solid solutions. The VESTA (visualization for electronic and structural analysis) software package [48] was employed to extract the planar charge density contours from the first-principle calculations.

3. Results and discussions

The addition of solute atoms can alter the plastic behavior of Mg by either reducing or increasing the plastic anisotropy based on the chemical and elastic interactions between the solute atom and the surrounding host lattice [47,49,50]. To analyze the effects of various solute atoms (Li, Sn, Zn, Zr and Ce) in Mg, the ideal shear resistance across different slip systems (Table S3) was quantified using the computational framework discussed in the earlier section. Furthermore, the suitability of the various solute atoms to provide strengthening/softening, due to their electronic structure, atomic radii and electronegativity, was thoroughly explored.

In the case of pure Mg, the ideal shear resistance along the basal partial $(\{0001\})\langle 10\bar{1}0\rangle$, prismatic $\left(\{10\bar{1}0\}\langle 11\bar{2}0\rangle\right)$, pyramidal $\left(\{10\bar{1}1\}\langle 11\bar{2}0\rangle\right)$ and basal $(\{0001\})\langle 11\bar{2}0\rangle$ slip systems was 1.48, 1.58, 1.62, and 1.76 GPa, respectively, and were in agreement with previous findings [44] (Fig. 1a–d). Furthermore, the basal partial slip system displayed the least resistance to the shear deformation suggesting that it would be the most active slip system during plastic deformation of pure Mg. Note that throughout the paper, the $\left(\{0001\}\langle 11\bar{2}0\rangle\right)$ slip system is referred as "basal" and the $(\{0001\}\langle 10\bar{1}0\rangle)$ slip system as "basal partial".

In order to understand the mechanical behavior of Mg alloys, the shear stress-strain response of the various Mg-X solid solutions were compared to pure Mg along different slip systems. The addition of Ce or Zr in Mg was found to decrease the ideal shear resistance across different slip systems when compared to pure Mg (Fig. 1e–f). For instance, the addition of Ce was found to decrease the ideal shear resistance of the basal partial slip system from 1.48

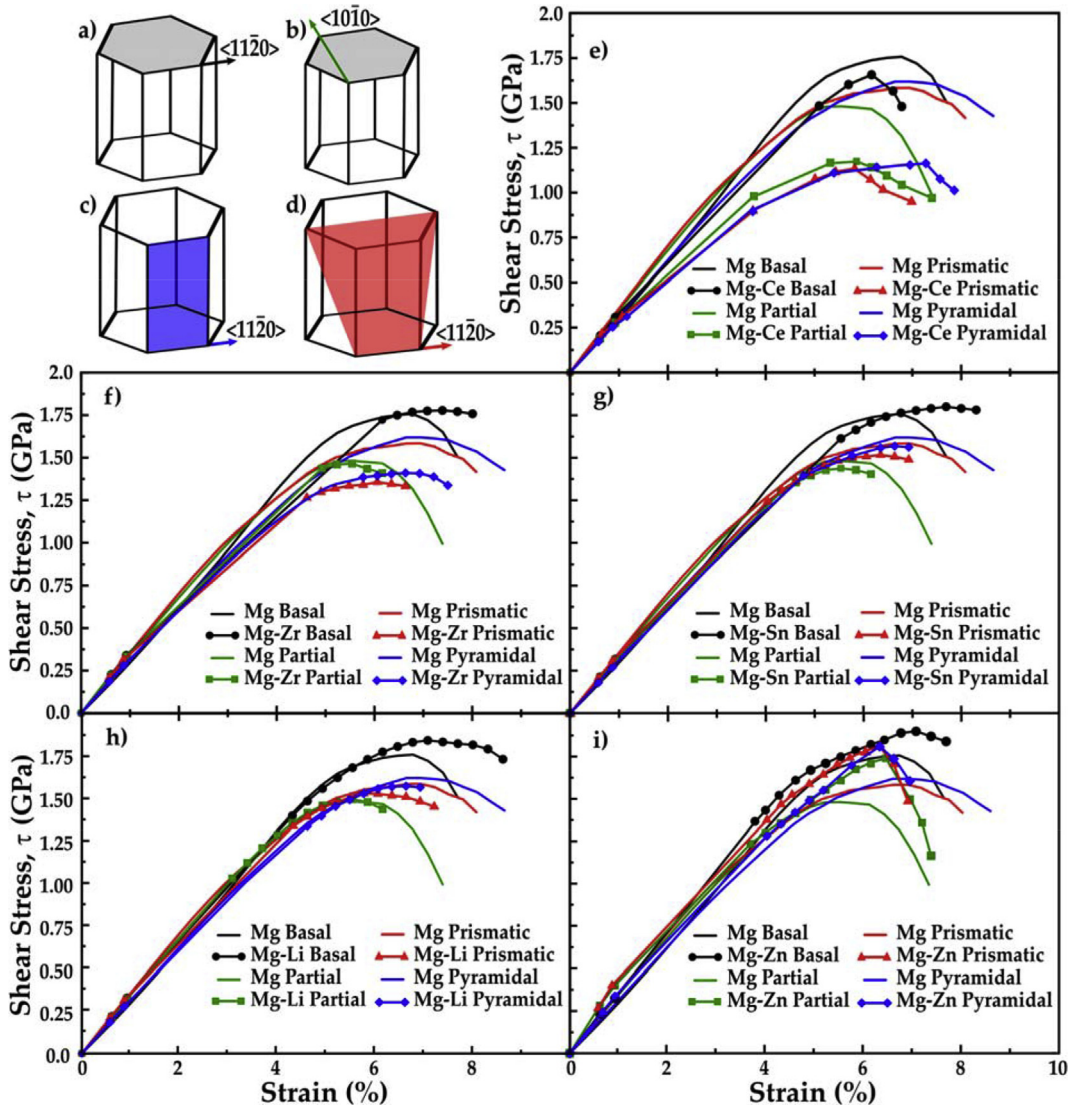

Fig. 1. The various slip systems in Mg (a) basal $\{0001\}\langle 11\bar{2}0\rangle$, (b) basal partial $\{0001\}\langle 10\bar{1}0\rangle$, (c) prismatic $\{10\bar{1}0\}\langle 11\bar{2}0\rangle$ and (d) pyramidal $\{10\bar{1}1\}\langle 11\bar{2}0\rangle$. The shear stress–strain response of e) Mg-Ce, f) Mg-Zr, g) Mg-Sn, h) Mg-Li and i) Mg-Zn solid solution, respectively, compared to pure Mg along different slip systems. Note: the basal full dislocation is referred to as "basal" and the basal partial dislocation as "basal partial".

to 1.17 GPa; a similar decrease in ideal shear resistance, varying from 5 to 28%, was observed along the other slip systems in the Mg-Ce solid solution. Furthermore, the least resistant slip system changed from the basal partial to the prismatic slip system, due to the addition of Ce or Zr (Fig. 1e–f). In other words, the addition of certain solutes (Ce and Zr) in Mg can alter the macroscopic plastic behavior, thereby improving the formability of Mg alloys.

Based on the promising findings of solute addition in Mg, the effect of other substitutional solute atoms, such as Sn, Li, and Zn were further explored. The addition of Sn led to a small increase in ideal shear resistance from 1.76 GPa to 1.80 GPa along the basal slip system when compared to the pure Mg (Fig. 1g). However, a negligible effect on the ideal shear resistance was observed along the remaining slip systems (basal partial, prismatic and pyramidal) of Mg-Sn alloy as compared to pure Mg with the basal partial still as the least shear resistant slip system (Fig. 1g). The addition of a Li atom resulted in a small increase and decrease in the ideal shear resistance along different slip systems of Mg-Li solid solution (Fig. 1h), but the basal partial remained the least shear resistant slip system in the Mg-Li solid solution (Fig. 1h). The solution strengthening of Mg by Zn solute atom was observed as the ideal shear resistance increased to 1.74 GPa along the basal partial slip system, when compared to pure Mg (1.48 GPa) (Fig. 1i). Although, the ideal shear resistance increased for all slip systems upon addition of Zn, the basal partial is still the least shear resistant slip system in Mg alloys (Fig. 1i). Hence, the basal partial was still the most dominant slip system and no additional slip systems are activated in Mg upon addition of Sn, Li and Zn solute atoms. Furthermore, significant strengthening is observed in the Mg-Zn solid solution while, the strengthening effects upon Sn or Li addition in Mg were found to be comparatively smaller.

The addition of a solute atom changes the surrounding atomic interactions and thus the examination of electronic structure can aid in understanding the strengthening/softening behavior in solution strengthened Mg alloys. Therefore, in this work, the effect of solute addition on the electronic density of states and valence charge transfer was examined to gain a deeper insight into the underlying factors influencing the ideal shear resistance of Mg-X solid solutions. The total electronic DOS for the various Mg solid solutions was obtained for the unstrained configuration (Fig. 2).

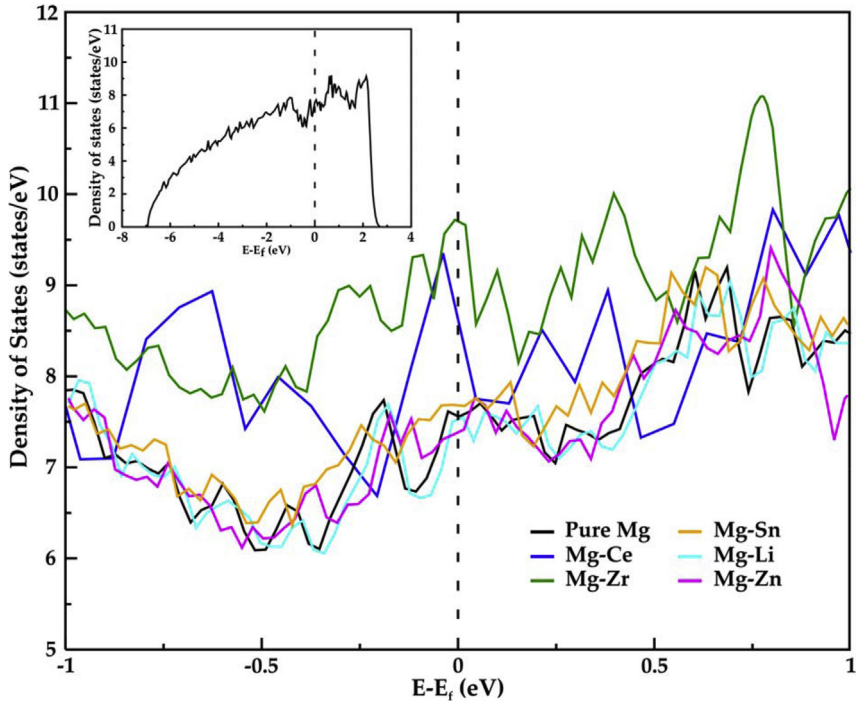

Fig. 2. Total density of states for pure Mg, Mg-Ce, Mg-Zr, Mg-Sn, Mg-Li and Mg-Zn solid solutions, respectively, near the Fermi level (0 eV).

The addition of Ce increased the DOS and decreased the electronic structure stability of the solid solution near the Fermi level due to the interaction of 5d states of Ce with 2*p* states of Mg. In Mg-Zr solid solution, the DOS curve was found to have a significant increase near the Fermi level due to the interaction of partially filled 4d states of Zr with 2*p* states of Mg. The *d* orbital contribution from Ce and Zr solute atoms was found to cause an increase in the total DOS of Mg at the Fermi level thereby decreasing the electronic structure stability (i.e., stability of electronic structure of atom) of Mg-Ce and Mg-Zr solid solutions. On the other hand, addition of Sn, Li and Zn had negligible changes in the DOS of the solution strengthened alloy (Fig. 2b). Therefore, suggesting a negligible effect on the overall bonding between the atoms in Mg-Sn solid solution and a very small increase in the bond strength between the atoms in Mg-Li and Mg-Zn solid solutions, respectively.

The material response to externally applied mechanical load is dependent on the strength of bonding between the atoms. To this end, the valence charge transfer in the various Mg solid solutions was quantified to provide insights into the redistribution of charge and strength of bonds. Furthermore, when the charge distribution is non-spherical/anisotropic the bonding between the atoms is directional leading to an increased ability to withstand shear deformation; whereas, the shear deformation occurs easily and the ideal shear resistance decreases when the charge density distribution is spherical/isotropic [51]. Therefore, it is crucial to understand the changes in valence charge distribution upon addition of solute atoms on the ideal shear resistance of different slip systems. The valence charge density difference was calculated as:

$$\Delta\rho = \rho_{Mg-X} - \rho_{Mg} - \rho_X \quad (1)$$

where $\Delta\rho$ represents the overall valence charge density difference in the solid solution while shearing, ρ_{Mg-X} is symbolic of the valence charge density of the Mg-X solid solution, and ρ_{Mg} and ρ_X represent the valence charge density of non-interacting isolated Mg atoms and an X atom, respectively [52]. The solute addition had a localized effect in its surrounding region while the Mg-Mg bonding in the solid solution is unaffected, since neither charge redistribution nor charge transfer occurs in the bulk metal. Fig. 3 shows the planar charge density on the 1) basal, 2) basal partial, 3) prismatic and 4) pyramidal slip systems. The solute atom is present in the center, represented by a white circle and surrounded by the Mg atoms at the corners of each square shaped charge density contour.

The valence charge transfer takes place from the Ce atom out towards the surrounding Mg atoms (Fig. 3a). The charge is densely distributed on the basal slip system as compared to the other three slip systems in Mg-Ce solid solution; hence, the bonding between the atoms is strong and explains the ability of the basal slip system to withstand maximum shear deformation (Fig. 3a). In Mg-Zr solid solution, the charge accumulated around the Zr atom is very small since Mg and Zr have similar electronegativity and atomic radius. The charge is densely distributed on the basal and basal partial slip systems as compared to the non-basal slip systems (Fig. 3b). The spherical charge density distribution on the prismatic slip system corresponds to weak bonding between the atoms, followed by the pyramidal slip system, thereby explaining the observed ideal shear resistance behavior across various slip systems of Mg-Zr solid solution (Fig. 3b). In the Mg-Sn solid solution, the valence *5p* electrons can contribute to the charge density between Sn and surrounding Mg atoms (Fig. 3c). Anisotropic charge distribution on the basal and the basal partial slip systems contributes to the bond strength as compared to the prismatic and pyramidal slip systems with a uniform (spherical) charge density (Fig. 3c). Similarly, the charge density distribution across the various slip systems illustrates the observed ideal shear resistance behavior of Mg-Li solid solution (Fig. 3d). In the Mg-Zn solid solution, the anisotropic charge density distribution due to the directional spreading of *d* orbital electrons can give rise to the high bond strength between the atoms and support the observed increase in the ideal shear resistance (Fig. 3e).

Next, the binding energy for the various solute atoms was subdivided into the mechanical (structural) and chemical (bonding)

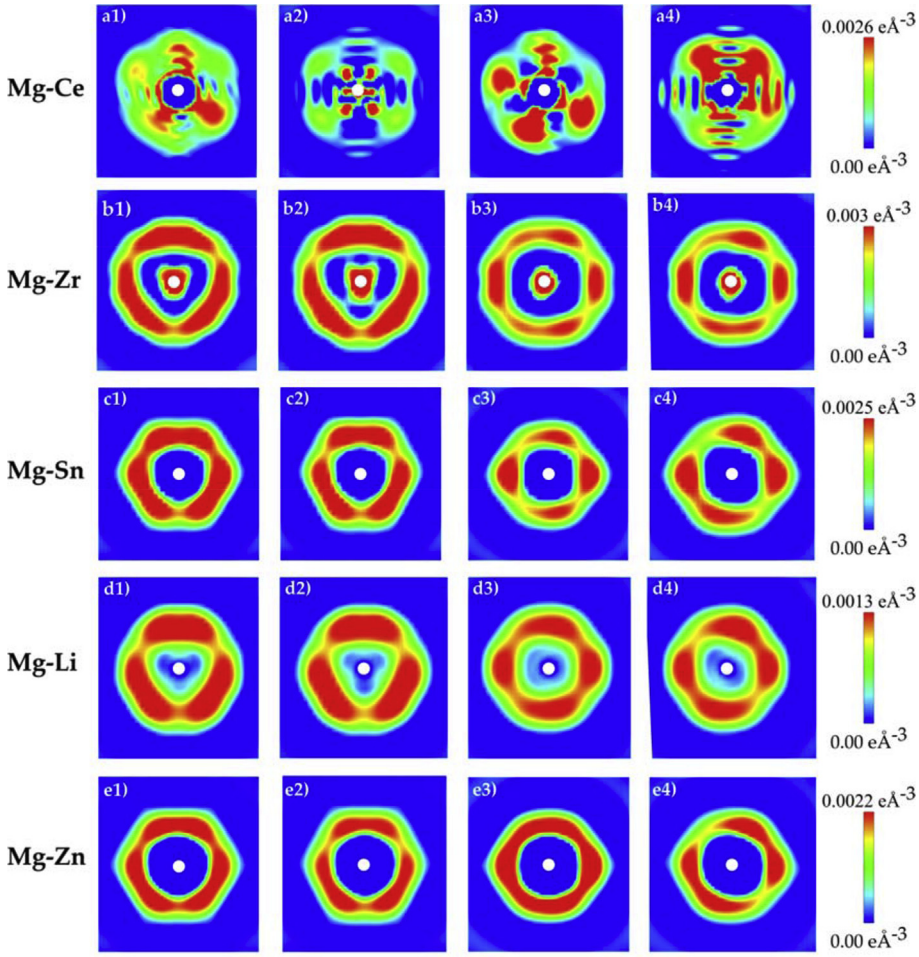

Fig. 3. The planar charge density (eÅ^{-3}) on the 1) basal, 2) basal partial, 3) prismatic and 4) pyramidal slip system around the solute atom at the corresponding maximum shear stress in the a) Mg-Ce, b) Mg-Zr, c), Mg-Sn, d) Mg-Li and e) Mg-Zn solid solutions. The solute atom is present in the center represented by a white circle and surrounded by the Mg atoms at the corners of each square shaped charge density contour.

contributions based on the formulae presented in Refs. [46,53,54]. The interactions between the solute atom and the host atoms were considered by the chemical contribution (E^c). Therefore, E^c is the energy difference between the relaxed structure containing the solute atom and the same structure where the solute atom has been removed without any subsequent relaxation of the host lattice, plus the cohesive energy of the solute atom (E_X):

$$E^c = E_{Mg_{107}X_1} - E_{Mg_{107}} - E_X \quad (2)$$

where, $E_{Mg_{107}X_1}$ is the energy of the relaxed host lattice with a solute atom and $E_{Mg_{107}}$ is the unrelaxed energy of the same structure after the removal of the solute atom. Similarly, the mechanical contribution (E^m) is accounted for by quantifying the energetics associated with the volumetric relaxation of the host lattice in the vicinity of the solute atom [54]. For a substitutional solute atom, the mechanical contribution depends on the energy change upon removal of a host metal atom from the lattice site and addition of the solute atom at the corresponding vacant site. Thus, mechanical contributions were computed by subtracting the energy of the relaxed host lattice after removing the solute atom without subsequent relaxation, from the relaxed energy of the host lattice:

$$E^m = E_{Mg_{107}} - 107 * E_{Mg} \quad (3)$$

where, E_{Mg} is the cohesive energy of the relaxed Mg atom. Table 1 shows the changes in mechanical and chemical contribution upon substitution of different solute atoms to pure Mg.

In the Mg-Ce solid solution, large volumetric distortions were introduced due to the much larger atomic size of Ce (185 p.m.) when compared to Mg (150 p.m.). This behavior was accurately captured by the positive mechanical contribution (Table 1). Further, a positive E^c value for Mg-Ce solid solution indicates strong chemical interactions between valence electrons of Ce and Mg which implies a decrease in strength of the bond between the host atoms. Therefore, the addition of Ce causes both volumetric distortions and weakening of bonds between the host atoms, which can aid in explaining the large decrease in the ideal shear resistance. In Mg-Zr solid solution, the lattice distortions dominate,

Table 1
The mechanical and chemical contributions to the binding energy (in eV) due to addition of different solute atoms in Mg.

	E^m (eV)	E^c (eV)
Mg-Ce	1.79	0.66
Mg-Zr	−0.64	−0.52
Mg-Sn	−0.61	−1.23
Mg-Li	−0.65	−0.96
Mg-Zn	−1.62	−0.84

since Zr (155 p.m.) has bigger atomic radii than Mg, thereby explaining the decrease in the ideal shear resistance of Mg-Zr solid solution. In Mg-Sn and Mg-Li solid solutions respectively, the bond strengthening between the host atoms dominates the volumetric distortions since the atomic radius of Sn (145 p.m.) and Li (145 p.m.) are slightly smaller as compared to Mg. The atomic radii of Zn (135 p.m.) is smaller than the atomic radii of Mg which may lead to volumetric contraction and increase the strength of bonds between the Zn and Mg atoms. Thus, even though Mg-Zn solid solution deviated from the trend in Table 1, the lattice contraction promotes bond strengthening between the atoms thereby increasing the ideal shear resistance of Mg-Zn solid solution. Thus, the strengthening/softening effect of solute atoms is partially dependent on the atomic radii of the solute atom and the host atoms.

In order to assess the effect of different solute atoms on the plastic anisotropy of Mg alloys, the activity of various slip systems when subjected to uniaxial tension along a wide range of crystallographic directions is discussed. The Schmid's law [55] and the ideal shear resistance across various slip systems (τ_{crss}) were employed to quantify the tensile stress required to activate the desired slip system. The experimentally calculated critical resolved shear stress is influenced by various defects like vacancies, dislocations etc. present in the crystal; however, in a defect free crystal, the critical resolved shear stress corresponds to the ideal shear resistance of the given slip system. Thus, the tensile stress required to activate the *i*th slip system (σ_y^i) for a given loading direction [uvw] is calculated as:

$$\sigma_y^{i,[uvw]} = \tau_{crss}^i \Big/ \cos\phi^{i,[uvw]} \cos\lambda^{i,[uvw]} \tag{4}$$

where, ϕ is the angle between the slip plane normal and loading axis and λ is the angle between the slip direction and loading axis. The slip system with the lowest tensile stress ($\sigma_y^{i,[uvw]}$) is the most active slip system for a given loading direction.

Next, the tensile stress ratio was defined as the ratio between the tensile stresses required for activating the basal partial slip system and the most active slip system (other than basal partial) for a given loading direction. For example, if the prismatic slip system is the most active for a given direction then the tensile stress ratio ($\sigma_{Mg}^{bp,[uvw]}/\sigma_{Mg}^{pr,[uvw]}$) is greater than 1. Whereas; for a given loading direction if the basal partial is the most active slip system and pyramidal is the second most active slip system, then the tensile stress ratio, $\sigma_{Mg}^{bp,[uvw]}/\sigma_{Mg}^{py,[uvw]}$ is less than 1.

The tensile stress ratios across the various tensile loading directions were mapped onto the stereographic triangle appropriate for HCP metals where the corners of the triangle represent $\langle 10\bar{1}0\rangle$, $\langle 11\bar{2}0\rangle$ and $\langle 0001\rangle$ loading orientations (Fig. 4). In Fig. 4, the black and white contour lines correspond to the tensile stress ratio of 1 for pure Mg and a solution strengthened Mg alloy, respectively. The region I in Fig. 4 corresponds to the loading directions where the basal partial system was active. Similarly, the region II and III on the stereographic triangle (Fig. 4) correspond to the loading directions where the preferential slip systems were prismatic and pyramidal.

The ideal shear resistance calculations for Mg-Y and Mg-Al solid solutions from our previous work [17] are also included to obtain an extensive understanding of the effect of various solutes on the plasticity of Mg alloys. The addition of Ce, Y and Zr to Mg was found to increase the fraction of loading orientations that favored non-basal slip systems, especially the prismatic (region II) and pyramidal (region III) slip systems as compared to pure Mg (Fig. 4a–d). In other words, the addition of Ce, Y and Zr increases the activation possibilities for non-basal slip systems and indicates a decrease in plastic anisotropy across a wide range of tensile loading orientations. On the other hand, the addition of Li, Al, Sn and Zn had negligible effects on the fraction of tensile loading orientations that favor non-basal slip systems (Fig. 4e–h). Therefore, the plastic anisotropy would remain largely unaffected for Mg alloys solute strengthened with Li, Al, Sn and Zn. Thus, the solid solution softening of Mg alloys with certain solutes (Ce, Y and Zr) can alter the ideal shear resistance behavior across the various slip systems, thereby decreasing the plastic anisotropy at the macroscopic level (Fig. 4b–d). In past experimental studies [56–58], Mg-Y and Mg-Ce alloys were found to have higher ductility and lower strength as compared to pure Mg due to the weakening of basal texture and formation of non-basal texture components in the alloys. Therefore, these alloys (Mg-Ce, Mg-Y and Mg-Zr) can deform with more active dislocation slip systems as compared to pure Mg which is in agreement with our ideal shear resistance calculations. The ideal shear resistance calculations also capture the experimentally observed solid solution strengthening in Mg-Zn alloys, due to increased strength and lower ductility, as compared to pure Mg [59–61]. Similarly, the increase in strength of Mg-Li and Mg-Al alloys observed in the experimental studies is also in good agreement with ideal shear resistance calculations [62,63] along with increased ductility in Mg-Al alloys due to softening of prismatic planes [64]. Thus, the systematic work in this study aids in the identification of solute atoms that can lead to an increase/decrease in the plastic anisotropy of Mg alloys. However, studying the dislocation behavior, grain boundary deformation and interaction of solute atoms with various defects are critical to obtain a comprehensive understanding about the deformation behavior of various Mg alloys.

Physically, the theoretical shear strength signifies the stress required to shear atomic planes in an ideal crystal. Frenkel [65] proposed a sinusoidal relationship between the shear stress and displacement to understand the microscopic plastic deformation process. Furthermore, Rice [66] utilized this relationship to estimate the unstable stacking fault energy as the area under the shear stress versus displacement curve. The criteria to predict the dominant slip system in a particular metal based on the stacking fault energy and elastic properties was first proposed by Legrand [67]. This parameter was further extended by Moitra et al. [47] to determine the effect of solute atoms in Mg with reference to pure Mg. However, the formability parameter in Ref. [47] is a function of the stacking fault energies and does not include the elastic properties. Thus, the proposed shearability parameter provides quantitative insights into the plastic behavior of Mg alloys by considering the ideal shear resistance and elastic properties for different slip systems. The shearability parameter (R_τ) is defined as:

$$R_\tau = [\tau_{BP}/C_{BP}]/[\tau_P/C_P] \tag{5}$$

where, τ_{BP} and C_{BP} are the ideal shear resistance and the shear modulus along the basal partial slip system, respectively, and τ_P and C_P are the ideal shear resistance and shear modulus along the investigated slip system. The shearability parameter (R_τ) serves as an index of the solute strengthening/softening behavior for a particular slip system when compared to the primary slip system (i.e. basal partial for Mg). For instance, a R_τ value greater than or close to 1 indicates a higher probability of activation for the non-basal slip systems. For example, in pure Mg the R_τ value was 0.90 for the prismatic slip system, but the addition of Ce increased the R_τ value to 1.15. The additions of a Ce, Y and Zr solute atom, were respectively found to increase the shearability parameter for non-basal slip systems (specifically the prismatic slip system) when compared to pure Mg (Fig. 5); thus, these solutes would improve

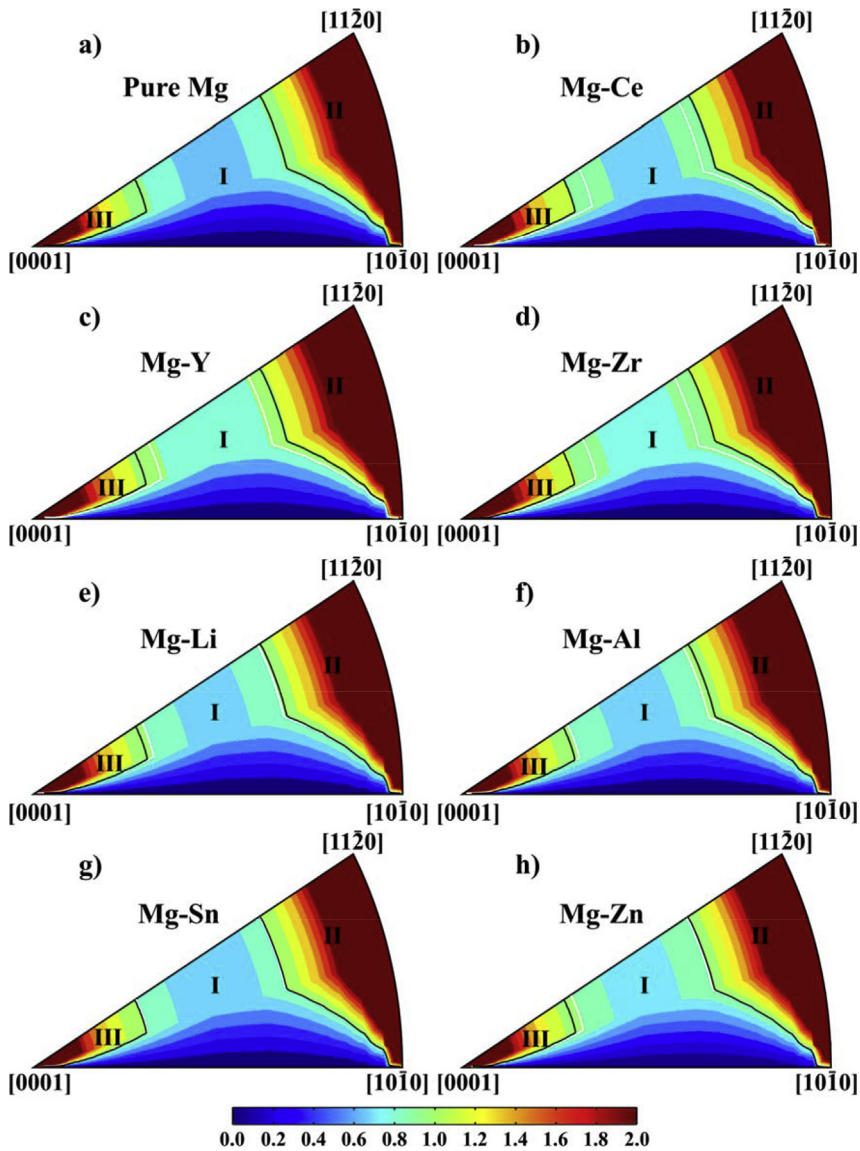

Fig. 4. Orientation dependence of different slip systems, (I) basal partial, (II) prismatic and (III) pyramidal, with respect to the basal partial slip system in a) pure Mg, b) Mg-Ce, c) Mg-Y, d) Mg-Zr, e) Mg-Li, f) Mg-Al, g) Mg-Sn and h) Mg-Zn solid solutions when subjected to uniaxial tension along different orientations. The addition of Ce, Y and Zr solute atoms to Mg, respectively increases the number of loading directions for which prismatic and pyramidal slip systems become active.

the macroscopic formability of Mg alloys. On the other hand, the basal partial slip system was the most preferred with the addition of Sn, Li, Al and Zn suggesting a negative impact on the formability of Mg alloys (Fig. 5). These observations from our theoretical calculations are found to be consistent with the experimentally and theoretically reported behavior, as observed in Refs. [18,41,45,47].

4. Conclusion

A comprehensive first-principles study was carried out to understand the implications of solute addition on the plastic deformation of Mg alloys. Specifically, the effect of different solutes (Li, Zn, Zr, Sn and Ce) on the ideal shear resistance along the various slip systems in Mg was quantified. Furthermore, the electronic structure (DOS), nature of bonding (charge density contours) and estimation of the segregation behavior revealed the underpinning reasons for the observed shear response of the various Mg alloys. The following are the key findings of this work:

1. The addition of Ce and Zr atoms decreased the ideal shear resistance along different slip systems of Mg alloys. Furthermore, the most active slip system changed from the basal partial to the prismatic slip system leading to the observed softening upon addition of Ce and Zr solute atoms to Mg. On the other hand, for the concentration studied here, the addition of Sn, Li and Zn atoms had negligible effect on the plastic anisotropy of Mg as the basal partial remained the most active slip system.
2. The DOS for the various Mg alloys revealed that the strong interactions of the *d* orbitals of Ce and Zr with *p* states of Mg contributed to a decrease in electronic structure stability and bond weakening that resulted in the reduction of ideal shear resistance across the various slip systems of Mg alloys.
3. The fraction of loading orientations that exhibited a preference for non-basal slip systems increased with the addition of Ce, Y and Zr in Mg, indicating a decrease in the plastic anisotropy of Mg alloys. On the other hand, the addition of Li, Sn, Al and Zn solute atoms had no significant effect on the plastic anisotropy

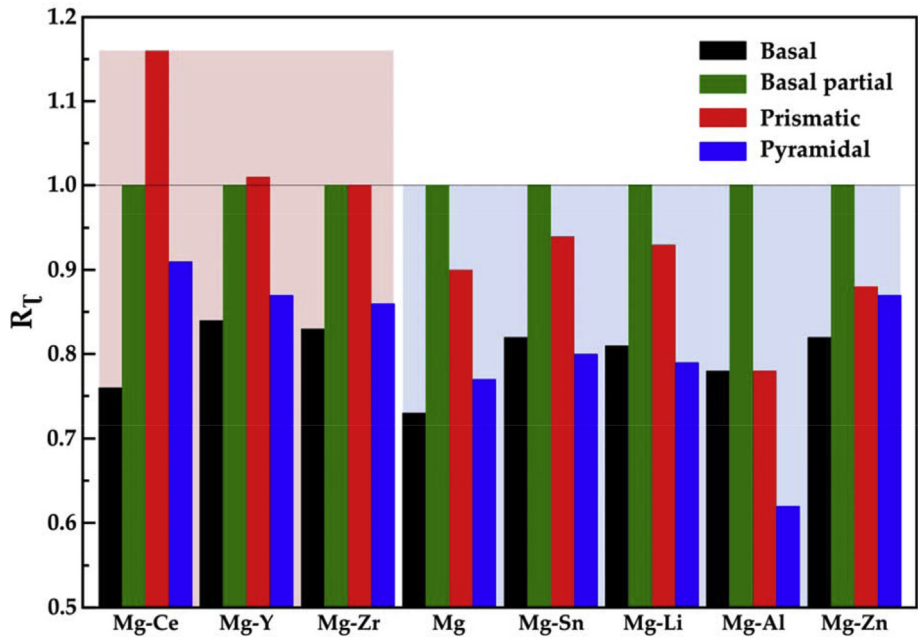

Fig. 5. Shearability parameter (R_τ) with respect to ideal shear resistance along various slip systems of Mg alloys.

(i.e. the activation of non-basal slip systems) and led to solid solution hardening of Mg alloys.

4. The calculated values of R_τ show that shearability increased along the prismatic slip system of Mg-Ce, Mg-Y and Mg-Zr solid solutions. On the other hand, the basal partial slip system had the maximum shearability for Mg-Sn, Mg-Li, Mg-Al and Mg-Zn solid solutions resulting in no improvement in the plastic anisotropy.

Overall, this work provides an insight into electronic and atomic basis on ideal shear resistance through the introduction of a shearability parameter, which considers the relative change in the critical shear resistance normalized with elastic properties in Mg binary alloys. However, systematic modeling efforts are necessary in the development of accurate continuum scale predictive capabilities that account for dislocation based plasticity, e.g., utilizing first-principles calculations to quantify the effect of solute addition on the glide behavior of dislocations along different slip systems. Nonetheless, in this work, a comprehensive ab-initio study was presented that can serve as a roadmap for efficient identification of feasible alloying elements to improve the formability of Mg alloys.

Acknowledgments

The authors are grateful for the financial support for this work from the National Science Foundation via award number 1463656. We also appreciate Fulton High Performance Computing at Arizona State University for enabling us to conduct our simulations.

Appendix A. Supplementary data

Supplementary data related to this article can be found at https://doi.org/10.1016/j.actamat.2018.05.014.

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